



MODELING THE PRECEDENCE EFFECT FOR PERCUSSIVE SOUNDS WITH DIFFERENT ATTACK TRANSIENTS

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ABSTRACT

In this investigation, a precedence model algorithm based on the Lindemann model [Lin:86a, Lin:86b] was used to simulate the precedence effect for impulsive percussive instrument sounds. Reference data was collected in a psychoacoustic experiment in which the echo thresholds for various percussive and non-percussive instrument sounds were determined for 12 listeners. In the headphone-based experiment, the lead was created with an interaural time difference (ITD) of 0.3 ms, while the lag was presented from the opposite side. Interstimulus interval (ISI, the delay between lead and lag) values of 0, 1, 2, 5, 10, 25, 50, and 100 ms were tested. The results of the experiment showed a large variation of the echo threshold. The smallest value of 7.50 ms was found for the castanet and the largest of 75.00 ms for a free-reed organ pipe. A regression analysis revealed that the duration of the attack transient, and the location of the harmonic centroid correlated best with the echo threshold. It is shown that the Lindemann model can be used to qualitatively explain the data.

INTRODUCTION

The human ability to localize a direct sound source in the presence of reflected sounds is well known as localization dominance due to the precedence effect, formerly also called “the law of the first wavefront.” The precedence effect is thought to be essential for humans to localize a sound source in reverberant environments. Our auditory system achieves this by suppressing the directional information regarding the reflections of the sound source (localization dominance). However, localization dominance is only observed, when the reflections arrive within a certain time interval, the inter-stimulus interval (ISI), after the primary source. If the time interval is shorter than 1 ms, the position of the auditory event is determined by both the primary source and the reflections. This effect is referred to as summing localization. If the time interval is larger than the echo threshold, the direct sound and reflections are perceived as two or more separate auditory events (echoes). Extended reviews on the precedence effect were written by Blauert [Bla:97], Litovsky et al. [Lit:99] and others.

In most investigations on the precedence effect, a single reflection and the direct sound source were used as stimuli. Here, the direct sound is usually referred to as the lead, while the reflection is called the lag. In early investigations, natural sounds were employed (e.g., [Haa:51]), where the echo threshold was determined to be 50 ms for speech and 80 ms for music. In most recent investigations click pairs or trains of clicks are utilized, which lead to lower echo thresholds than those found by [Haa:51]. To further understand how the echo threshold is affected by the duration, attack time and other parameters of the test stimuli, the authors recently conducted a psychoacoustic experiment in which the echo thresholds were determined for various percussive and non-percussive sounds [Sav:06a, Sav:06b]. The data of this experiment, which will be outlined in the following, were used for the model analysis that is described in this paper.

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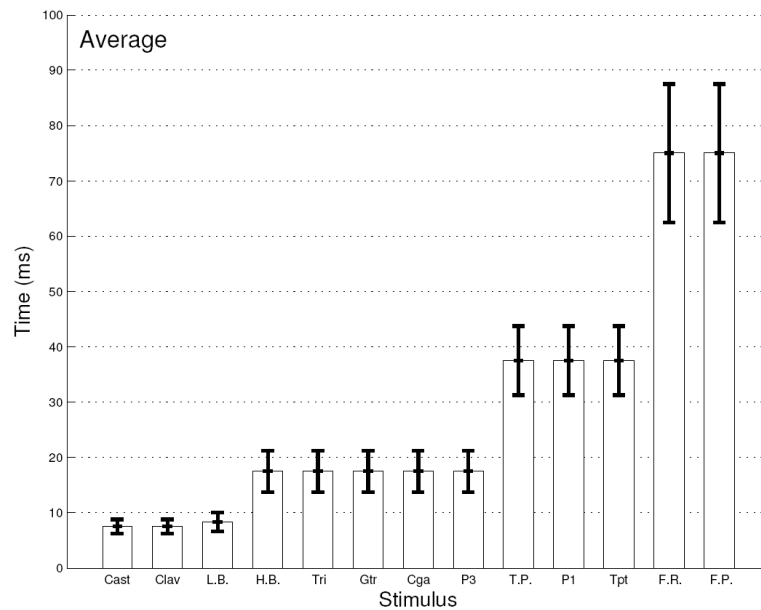


Figure 1: Echo Thresholds for various percussive and non-percussive instrument sounds: Cast=Castanet, Clav=Clave, L.B.=Low China Bell, H.B.=High China Bell, Tri=Triangle, Gtr=Guitar, Cga=Conga, P₃=Piano C₃, Tpt=Trumpet Organ Pipe, P₁=Piano C₁, F.R.=Free-Reed Organ Pipe, F.P.=Flue Pipe. The Echo Thresholds are the median of all listeners.

PSYCHOACOUSTIC EXPERIMENT

Methods

Eleven listeners (2 female, 9 male) participated in the experiment. Their ages ranged from 22 to 35 years. All listeners reported normal hearing. All musical instruments were recorded in the near-field in a low-reverberant environment with a resolution of at least 16 bit and 44.1-kHz sampling rate. The lead and lag stimuli varied by ISI values, while the ITD differed between +300 μ s and -300 μ s (approximately corresponding to a 45° angle of incidence). Half of the stimuli were presented with the right channel as the lead, and half were presented with the left channel as the lead. The ISI values of the experiment were randomly varied among the following values: 1, 2, 5, 10, 25, 50, 100 ms. Each ISI value was tested ten times. The stimuli were presented via headphones (Sennheiser HD-600). All recordings were then calibrated to a maximum of 65 dBA using a Head Acoustics Type HMS II dummy head. Each sound was calibrated and normalized by taking the average root means square (RMS) loudness as a reference. The listeners were asked to report whether they perceived one or more auditory event after each stimulus presentation.

Sound	Cast	Clav	F.P.	H.B.	L.B.	F.R.	Tri	T.P.
Echo Threshold (ms)	7.50	7.50	75.00	17.50	8.33	75.00	17.50	37.50
Attack (ms)	0.74	1.36	32.82	1.13	2.76	34.71	0.77	11.93
Decay (ms)	3.42	7.60	58.19	96.08	122.30	27.41	427.22	62.28
Duration (ms)	5.50	11.70	186.29	105.68	144.29	142.66	457.93	175.07
Centroid (FBI)	24.53	18.79	8.23	29.00	29.13	9.88	29.78	17.38

Sound	Guitar	Conga	Piano (C1)	Piano (C3)	Trumpet
Echo Threshold (ms)	17.50	17.50	37.50	17.50	37.50
Attack (ms)	5.97	4.85	12.45	12.38	95.63
Decay (ms)	148.93	66.16	205.77	205.55	117.04
Duration (ms)	188.68	84.32	319.92	272.71	366.60
Centroid (FBI)	5.89	8.30	10.29	9.36	19.03

TABLE 1: Physical parameters of the test stimuli.

Attack transient and decay time calculation methods

The attack transient time calculations involved a gammatone filter bank with 36 auditory bands at a sampling frequency of 44.1 kHz. After the filtering process, the Hilbert Transform was then used in each frequency band, and the attack transient time interval (ms) was calculated from the 20% to 80% onset thresholds. The decay was calculated using the same method as finding the attack, except calculating the 80% to 20% offset thresholds. The duration was calculated using the 20% onset threshold to the 20% offset threshold. Within the attack, decay and duration calculations, the spectral centroid was calculated according to [Bea:82].

Results

The psychoacoustic results are shown in Fig. 1. Each bar shows the 50% threshold as a median value over all listeners. The error bars depict the upper and lower quartiles. The stimuli with the lowest echo threshold values are the castanet and clave at 7.50 ms; the stimuli with the highest echo threshold values are the free-reed organ pipe and the flue pipe at 75.00 ms. In the debriefing, all listeners noted that the free-reed organ pipe and flue pipe stimuli were the most difficult to listen to when trying to distinguish the lag stimuli. Table 1 displays the physical parameters of the test stimuli.

Table 2 shows the results of the correlations between attack, decay, duration, spectral centroid and their log values and the measured echo thresholds. The analysis reveals that the logarithm of the attack time and the centroid correlate best with the echo threshold. The log of the attack transient times reveal a r -value of 0.74; meaning that the echo threshold increases with the attack transient time. Observing the p -values for the same correlation reveals a 0.0035 probability that the r -values are random. The correlation between the spectral centroid and the echo threshold reveals a negative linear relationship. With a r -value of -0.48 and a p -value of 0.14, this correlation shows that when the spectral centroid is low the echo threshold will be high (and vice versa).

Correlation	r	p
Attack & E.T.	0.52	0.070
Decay & E.T.	-0.19	0.54
Duration & E.T.	0.19	0.54
Centroid & E.T.	-0.48	0.097
log(Attack) & E.T.	0.74	0.0035
log(Decay) & E.T.	0.045	0.88
log(Duration) & E.T.	0.38	0.20
log(Centroid) & E.T.	-0.44	0.14

Table 2: Correlation between various physical parameters of the test sounds and the echo thresholds.

MODEL ANALYSIS

Model architecture

Lindemann introduced contralateral-inhibition elements (static inhibition) into a standard cross-correlation model, which allowed him to simulate the precedence effect for impulse sounds. The concept of the Lindemann model is described in [Lin:86a] in detail. Originally, Lindemann had only analyzed narrowband signals, and thus, no bandpass filter bank was required in his investigation. In our analysis, Basilar-membrane and hair-cell behavior are simulated with a gammatone-filter bank and a simple half-wave rectification which were implemented as shown in Fig. 2. The boxes *cue estimation* depict the cross correlation algorithms with the contralateral-inhibition elements. The gammatone-filter bank, which was described by Patterson *et al.* [Pat:95], consists, in this investigation, of 15 frequency bands from 180 Hz to 1670 Hz center frequency in steps of one ERB. The sampling frequency of the model was set at 40 kHz, and internal delays between $-1000 \mu\text{s}$ and $1000 \mu\text{s}$ were considered. After the half-wave rectification, the interaural cross correlation is estimated within each frequency band over the

whole stimuli duration. The constants of the inhibitory elements (which are explained in [Lin:86a]) were set to the following values: static inhibition constant $c_s=0.9$, dynamic inhibition constant $c_d=0.5$, fade-off time constant of the nonlinear low pass $T_{inh}=10$ ms.

One characteristic of the Lindemann model is that the effective degree of inhibition depends on the signal's amplitude. To avoid the degree of inhibition being much lower in those frequency bands with less signal energy, the signal's maximum in each band is scaled to one. After the half-wave rectification, the cross-correlation functions are computed and multiplied with the average power of the stimulus in the left and right channels measured in that frequency band. The cross-correlation functions are added up across frequency for the final binaural-activity maps.

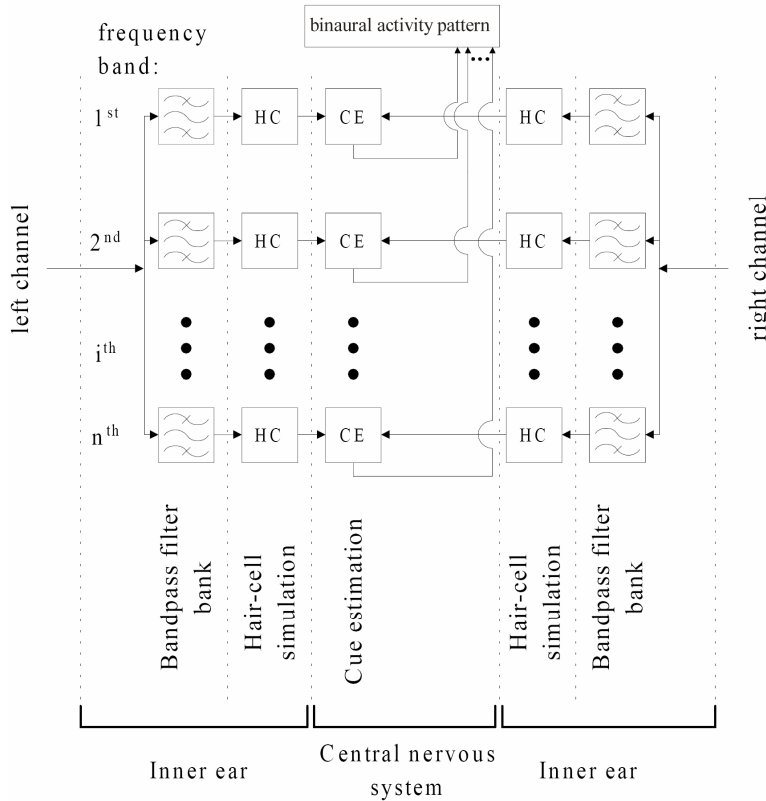


Figure 2: Model architecture.

Results

Figure 3 shows the results of the Lindemann model for the castanet, which was the tested stimulus with the shortest attack transient, duration, and echo threshold. The y-axis displayed the time course of the stimuli, the x-axis the internal delay. The z-axis shows the activities for the given time/internal delay bins. The figure depicts how the response for the lag develops with increasing ISI. The lag is not visible at an ISI of 2 ms (top left panel), but becomes noticeable at an ISI of 10 ms (bottom left panel). However, in this case the position of the lag is still in the direction of the lead (+300 μ s internal delay). The binaural response for lag appears at its presented ITD of -300 μ s when the ISI is adjusted to 25 ms. This value is slightly higher than the measured threshold of 7.5 ms.

In contrast to the castanet, a separate event for the lag is not visible for the flue pipe stimulus at an ISI of 25 ms (Fig. 4, top-left graph). The second event starts to develop at an ISI of 50 ms (Fig. 4, top-right graph), and is fully visible at an ISI of 100 ms (Fig. 4, bottom-left graph). The echo threshold was measured at 75 ms, and in the psychoacoustic experiment, an echo had not been perceived at 50 ms, while it had been detected in nearly all presentations at 100-ms ISI. One of the difficulties in determining whether a separate echo had been perceived or not, was

the observation that the auditory events dynamically changed their lateral position during the presentation. In this context it is noteworthy that only in the 100-ms condition the peak of the echo rises after it shifts to the internal delay of the lag. In the 50-ms condition, the peak shifts dynamically, but does not rise in amplitude afterwards. In fact, these dynamic shifts appear to be most prominent for the free-reed sample, and it was not surprising to us that the dynamic shift of the binaural image is more characteristic for this stimulus at 50-ms ISI (Fig. 4, bottom-right graph) compared to the flue-pipe case (Fig. 4, bottom-right graph).

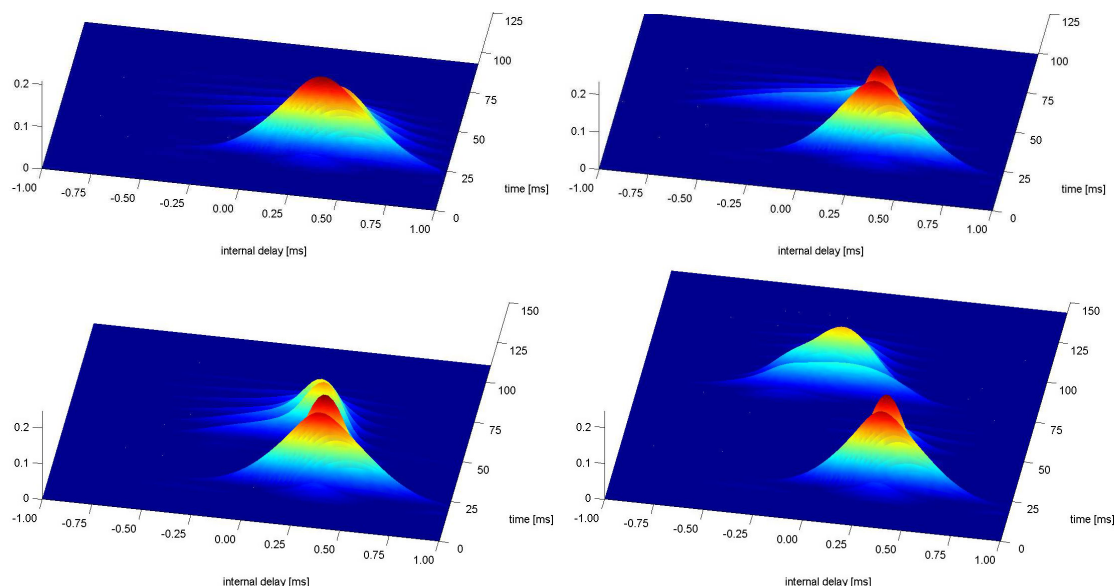


Figure 3: Binaural-activity map for a Castanet for different ISI values: 2 ms (top left), 5 ms (top right), 10 ms (bottom left), 25 ms (bottom right).

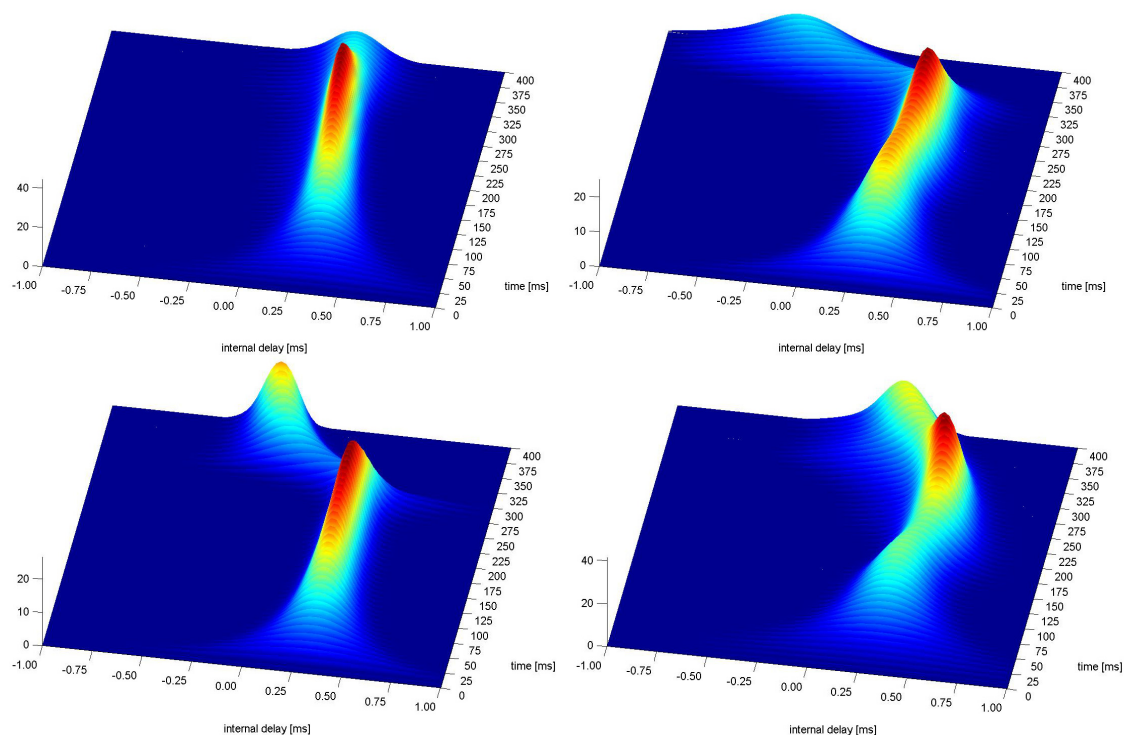


Figure 4: Binaural-activity map for a flue pipe and a free reed for different ISI values as noted – top-left: flue pipe, 25-ms ISI; top-right: flue pipe, 50-ms ISI; bottom-left: flue pipe, 100-ms ISI; bottom-right: free reed, 50-ms ISI.

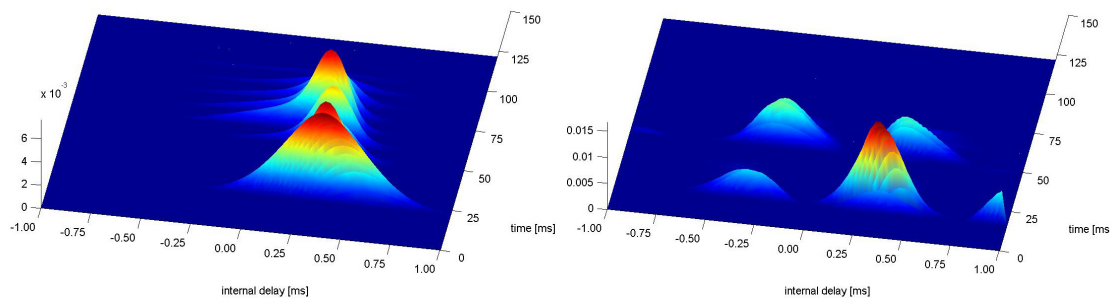


Figure 5: Binaural-activity map for a Castanet with an ISI value of 15 ms for a single frequency band. The left panel shows the data for the frequency band centered at 250 Hz, the right panel the same configuration for the frequency band centered at 1000 Hz.

Besides attack time, the location of the harmonic centroid appeared to be the most influential parameter to determine the echo threshold, and a higher centroid typically leads to a shorter echo threshold. This fact is easily explainable by the greater sluggishness of the auditory filters in the low-frequency range. Already Hartung and Trahiotis had pointed out how the sluggishness of the auditory periphery can be accounted to explain the precedence effect for impulsive stimuli. Compare the binaural-activity map for the castanet sample for the frequency band around 250 Hz to the same data in a higher frequency band. In the first case (Fig. 5, top-left) only one event appears at +300 μ s (which is extended in time), while for a center frequency of 1000 Hz, two events occur at the positions of both lead (+300 μ s) and lag (−300 μ s). Both events are clearly separated in time.

CONCLUSIONS

It has been shown qualitatively how the Lindemann model can be utilized to demonstrate how the echo threshold manifests for different impulsive sounds. In particular, it was shown how the duration of the attack transients of the stimuli affects the shape of the binaural-activity map, and why dynamic shifts in the lateral position are perceivable for stimuli with long attack transients. It was briefly shown how the frequency-dependent sluggishness of the auditory filters can be used to explain the influence of the centroid on the echo threshold. In future, we plan to extend this research to a quantitative analysis of the presented psychoacoustic data.

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